

Chapter 8: Kinetic Particle Model of Matter

Kinetic Particle Model of Matter (definition)

Kinetic Particle Model of Matter states that matter is made up of tiny particles that are in continuous motion.

Physical Properties of Solid, Liquid, Gas

Solid

1. Arrangement

→ Particles are very closely packed in an orderly manner, very little space between particles.

2. Movement

→ Particles rotate and vibrate about fixed positions.

3. Attractive Forces

→ Particles are held in position by strong intermolecular forces of attraction.

Liquid

1. Arrangement

→ Particles are closely packed in a disorderly manner, slightly further apart than in solids.

2. Movement

→ Particles slide past each other.

3. Attractive Forces

→ Particles are not held in fixed positions due to relatively weaker intermolecular forces of attraction.

Gas

1. Arrangement

→ Particles are spaced very far apart in a disorderly manner, mainly empty space between molecules.

2. Movement

→ Particles move quickly and randomly in any direction.

3. Attractive Forces

→ Particles are held together by very weak intermolecular forces of attraction.

Energy and Speed of Particles

Internal Energy of Particles

1. Internal Potential Store

→ changes during change of state of matter.

2. Internal Kinetic Store

→ changes during change of temperature.

Speed of Particles (Chapter 7)

$$KE = \frac{1}{2}mv^2$$

As KE increases, speed of particles increases

Higher temp = Higher speed of particles

Solid → Lowest KE → Lowest speed

Liquid → Low KE → Low speed

Gas → Highest KE → Highest speed

Explaining physical properties of matter using Kinetic Particle Model of Matter

1. Volume and Shape

Solid - Fixed shape & fixed volume

→ Arrangement + Attractive Forces

Liquid - Fixed volume, takes shape of container

→ Attractive forces: strong enough to prevent particles from leaving the substance.

→ Arrangement + Movement

Gas - No fixed shape and volume

→ Movement + Attractive Forces

2. compressibility - Arrangement

Solid & Liquid - Incompressible

Gas - compressible

3. Density

Solid - High density

→ Arrangement

→ There is a high number of particles per unit volume.

Liquid - Relatively high density

→ Arrangement

→ There is a relatively high number of particles per unit volume.

Gas - Low density

→ Arrangement

→ There is a low number of particles per unit volume.

Inter-Distance and Intermolecular Forces of Attraction

Intermolecular forces of attraction $\uparrow = \downarrow$ inter-distance
(vice-versa)

Solid → Strong intermolecular forces → smallest inter-distance

Liquid → Relatively weaker intermolecular forces → small inter-dist.

Gas → ^{very} weak intermolecular forces → large inter-distance

Brownian motion (definition)

Brownian motion is a phenomenon that provides evidence for the random and continuous motion of tiny particles.

Evidence of Brownian Motion

1. Pollen grains moving randomly and continuously in water
 2. Smoke particles moving randomly and continuously in air
- When tracing path of pollen/smoke, draw multiple random, straight arrows connecting to one another.

Motion of Molecules and Temperature

Temperature rises with ^{average} energy in internal kinetic store of particles and vice versa.

→ When substance is heated, energy is transferred to the kinetic store of the molecules. Energy in the kinetic store of the molecules increases.

→ Motion of molecules increases as temperature increases.

Motion of Molecules and Pressure (gas pressure)

Pressure increases with increase in speed of motion of molecules and vice versa. (specifically gas particles)

→ Gas particles are moving constantly in random direction.

→ When molecules collide with the inner walls of the container, they exert a force.

→ Since $\text{pressure} = \text{force} / \text{area}$, there will be pressure acted on the ^{inner} walls of container, known as gas pressure.

Pressure - Temperature of a Gas (constant volume)

The temperature and pressure of a fixed mass of gas are related.

→ When energy is transferred to gas through heating, temperature increases. This increases the average energy in the internal kinetic store of the particles, increasing the speed of particles.

→ Particles collide with ^{inner} walls more frequently with larger force.

→ Since $\text{pressure} = \text{force} / \text{area}$, when force $\uparrow$ but area remains constant, gas pressure in container increases.

Chapter 9: Thermal Processes

Thermal Equilibrium (definition)

Thermal equilibrium describes a state in which two objects have the same temperature and there is no net transfer of energy between them.

Energy transfer by heating

Energy transfer by heating always from a region of higher temperature to a lower temperature.

(stops at thermal equilibrium)

Conduction (definition)

Conduction is the process of energy transfer without any flow of the material medium. Thermal energy is transferred through the passing on of vibrational motion from one particle to another.

Conduction in Solids

1. Conduction in metals

Conduction takes place through vibrating molecules as well as free electrons.

- Small size & high speed of free electrons allow them to move easily and quickly.

- Transfer of thermal energy via free electrons is very fast, making metals good thermal conductors.

2. Conduction in non-metals

Conduction takes place through vibrating molecules.

- Thermal energy transferred through collisions of molecules.

Conduction in liquids and gases (fluids)

Poor thermal conductors compared to solids as particles are spaced further apart (less frequency of collisions of molecules)

Conduction - Terminologies

1. Energy transfer by conduction
2. Good/poor thermal conductor

Applications of Conduction

1. Good Thermal conductors - used where energy has to be quickly transferred by heating
 - Pans, Kettles, cooking utensils
2. Poor Thermal conductors - used to keep unwanted heat away/prevent heat loss
 - Double-glazed window, wool, fibreglass (trapped air = poor conductor)

Convection (definition)

Convection is the transfer of thermal energy by means of convection currents in a fluid, due to difference in density.

- cannot occur in solids

Convection currents - process

General process of convection currents - heating

1. Fluid at bottom gains thermal energy and expands, becomes less dense and rises.
2. Cold and denser fluid sinks to take the place of the rising fluid, and also gets heated up.
3. This sets up a circulation within the fluid.

(convection current)

This allows all parts of the fluid to gain thermal energy.

General process of convection currents - cooling

1. Fluid at top loses thermal energy and contracts, becomes denser and sinks.
2. Hotter and less dense fluid rises to take the place of the sinking fluid, and also gets cooled down.
3. This sets up a circulation within the fluid.
(convection current)
This allows all parts of the fluid to gain thermal energy.

Convection - Terminologies

1. Energy transfer by convection

Applications of Convection

1. Heating

- Kettles, Hot air balloons, hot water systems, heaters

2. Cooling

- Air-conditioners, refrigerators

Radiation (definition)

Radiation is the process of energy transfer by electromagnetic waves. Does not require a medium.
(can be done in vacuum)

Factors Affecting Rate of Radiation

1. Surface area

- The larger the surface area, the higher the rate of energy transfer by radiation.

2. Colour and texture of surface

- Black surface emits and absorbs radiation better than white surface.
- Rough surface emits and absorbs radiation better than smooth surface.

3. Surface temperature difference between surface and surrounding

- The larger the surface temperature difference, the higher the rate of energy transfer.

Radiation - Terminologies

1. Energy transfer by radiation

2. Good/Poor Emitter of Radiation

3. Good/Poor Absorber of Radiation

Applications of Radiation

1. Poor Radiators (Emitters & Absorbers)

- Aluminium foil, teapots (keep things warm)

- Walls painted white (keep things cold)

2. Good Radiators (Emitters & Absorbers)

- Dark clothes, solar panels (allow heating)

- Cooling fins of refrigerator (allow heat to be transferred to surrounding)